

- 6 (a) Fig. 6.1 shows an oxygen dissociation curve for adult human haemoglobin.

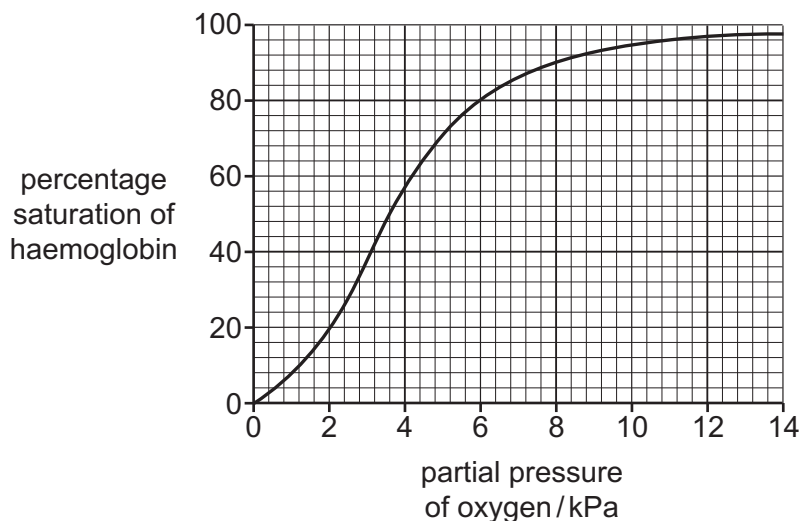


Fig. 6.1

An increase in the partial pressure of carbon dioxide ($p\text{CO}_2$) in respiring tissue causes the Bohr effect.

- (i) Sketch on Fig. 6.1 to show how the Bohr effect changes the oxygen dissociation curve. [1]
- (ii) Explain how an increase in $p\text{CO}_2$ produces the Bohr effect **and** state the benefit of this effect for the tissue.

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..... [3]

- (b) Carbon dioxide (CO_2) is transported across the cell surface membrane of the red blood cell using a different mechanism to the transport of hydrogen carbonate ions (HCO_3^-).

Name the different mechanisms of transport used for CO_2 and for HCO_3^- **and** explain why they are transported across the membrane by different mechanisms.

CO_2

HCO_3^-

explanation

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[4]

[Total: 8]